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(54) **CONNECTOR SET MATING METHOD FOR CONNECTING TWO PLUG CONNECTORS TO TWO SOCKET CONNECTORS AND A CONNECTOR SET USING THE SAME**

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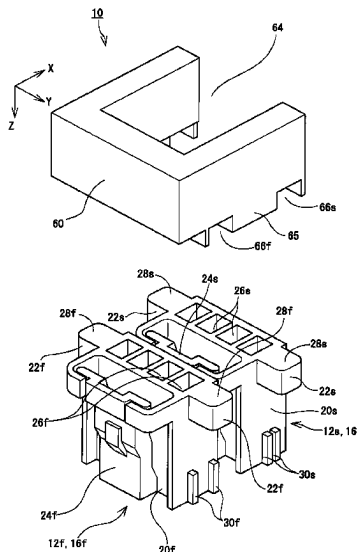
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(57) **ABSTRACT**

Each of two plug connectors includes a main body configured to retain a plurality of electric wires and a pair of push members protruding from both sides of the main body along a first direction. In connector mating operation, the two plug connectors are respectively inserted into associated receptacle connectors in a state where the two plug connectors are mechanically coupled or bound to a relative positional relationship by a jig in a position where the pair of push members in one of the two plug connectors are displaced from the pair of push members in the other of the two plug connectors along a second direction orthogonal to the first direction.

5 Claims, 9 Drawing Sheets



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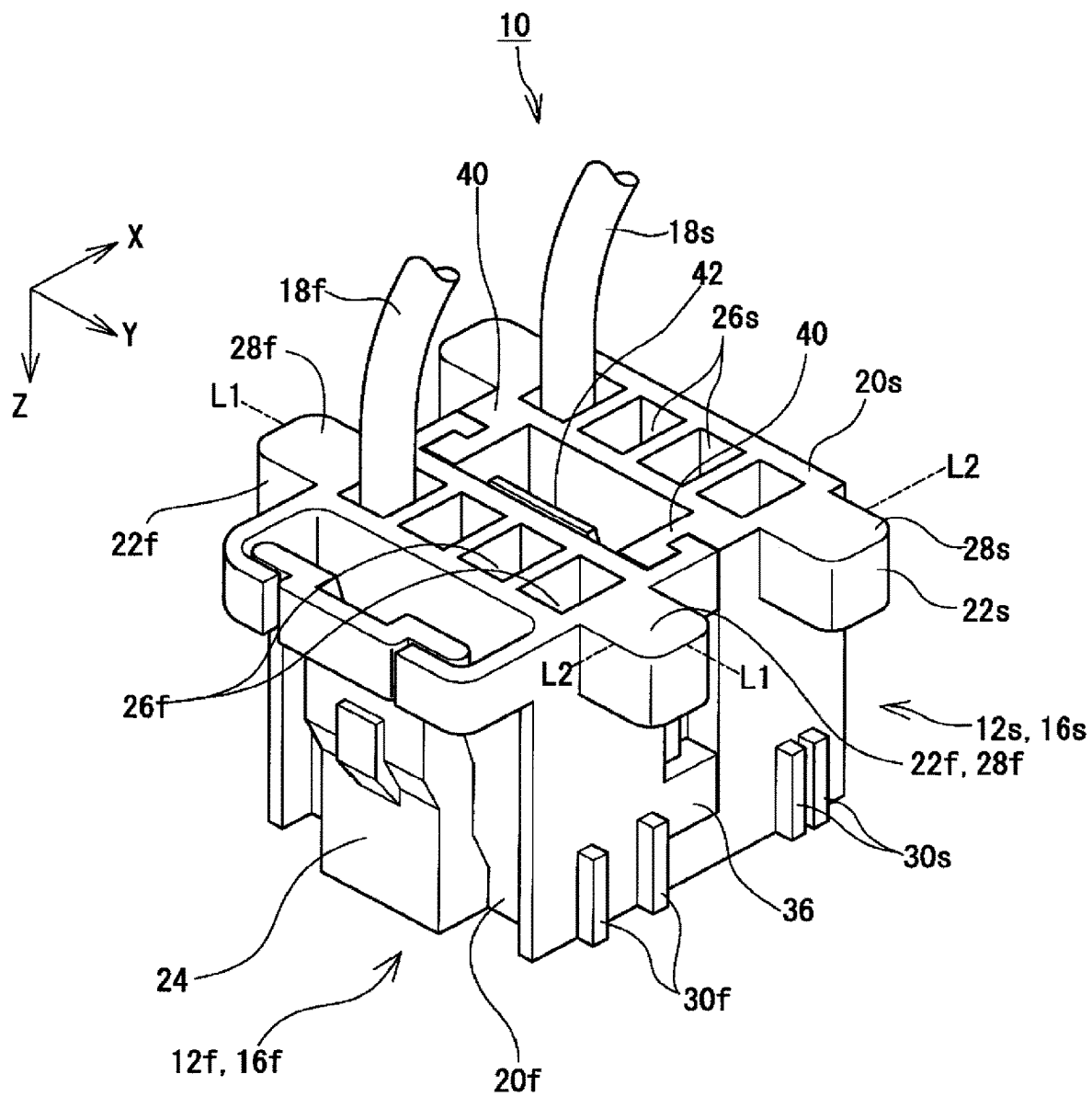


FIG. 1

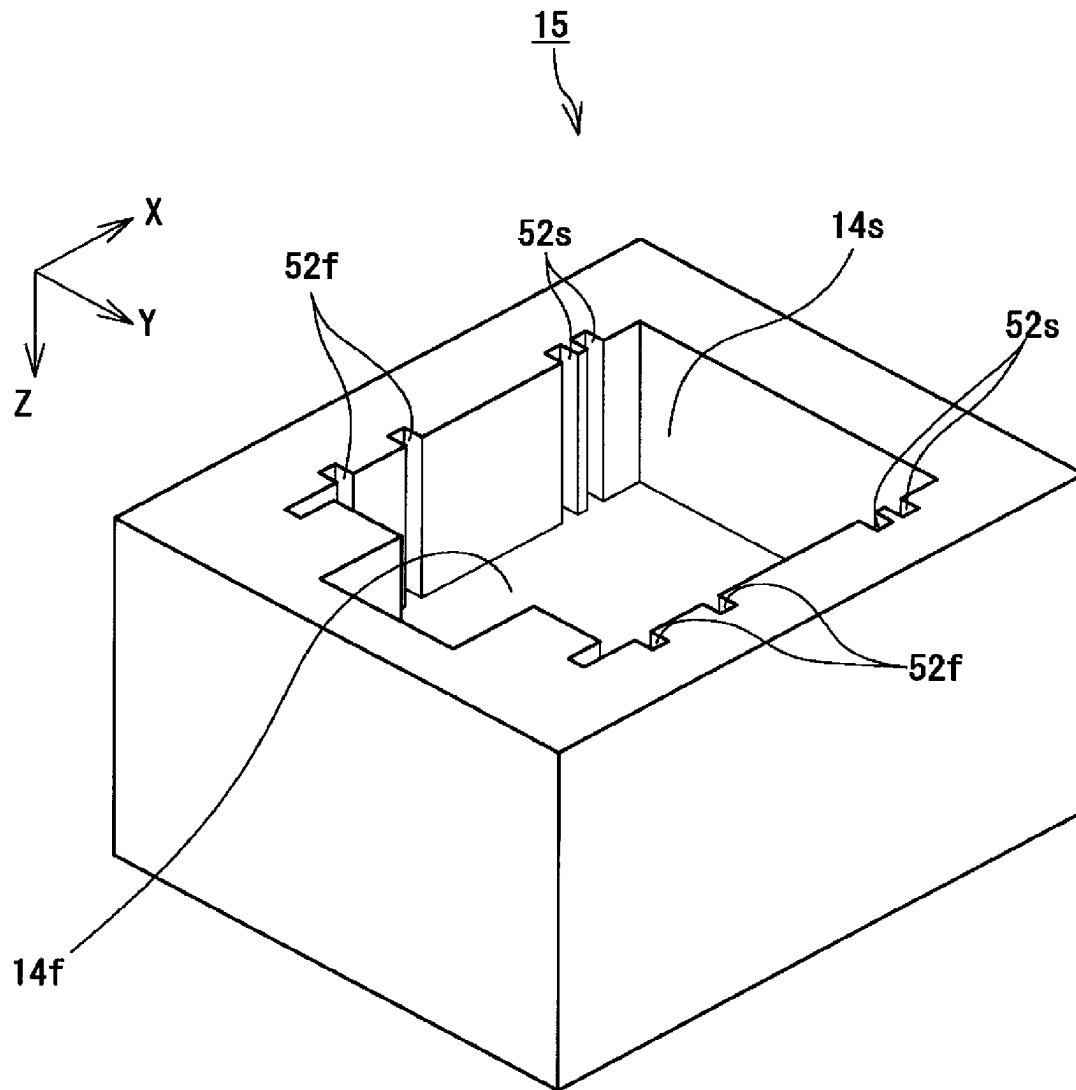


FIG. 2

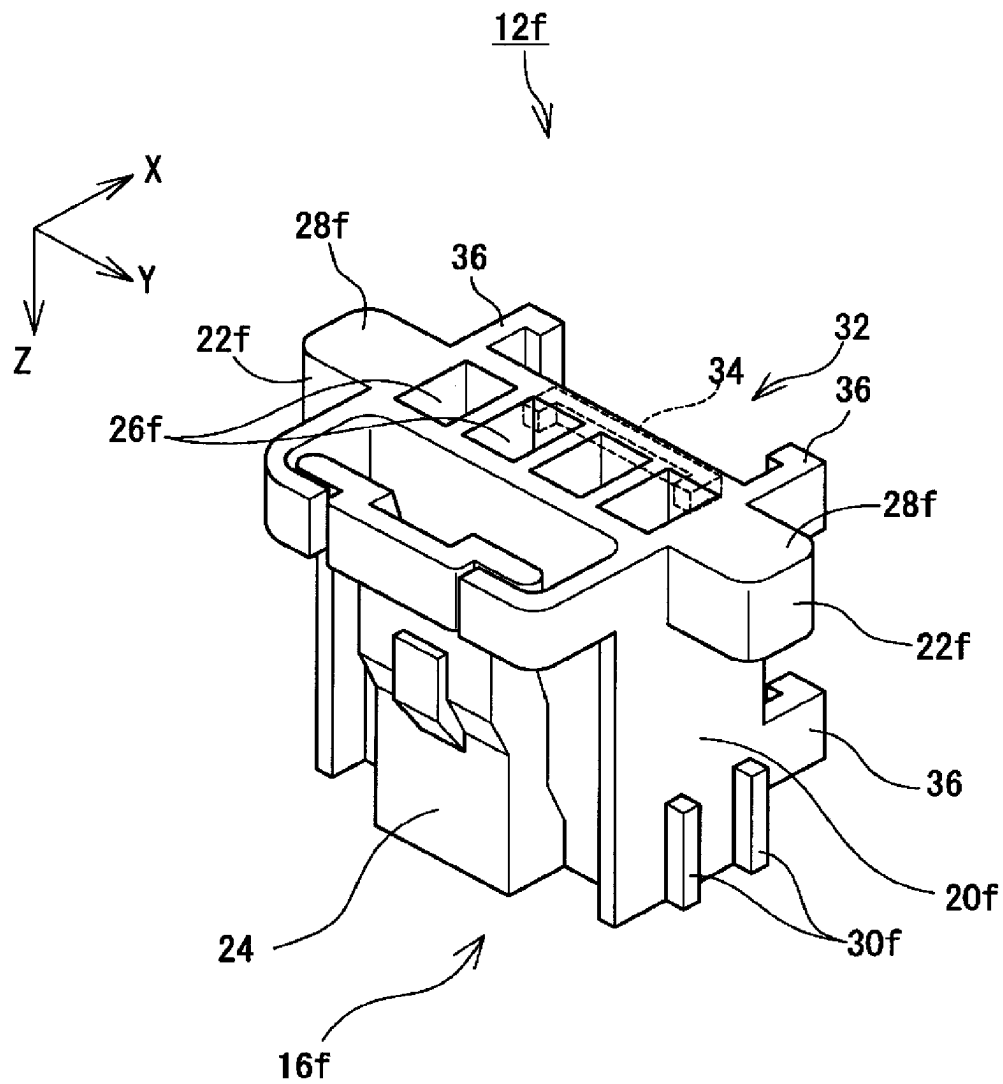


FIG. 3

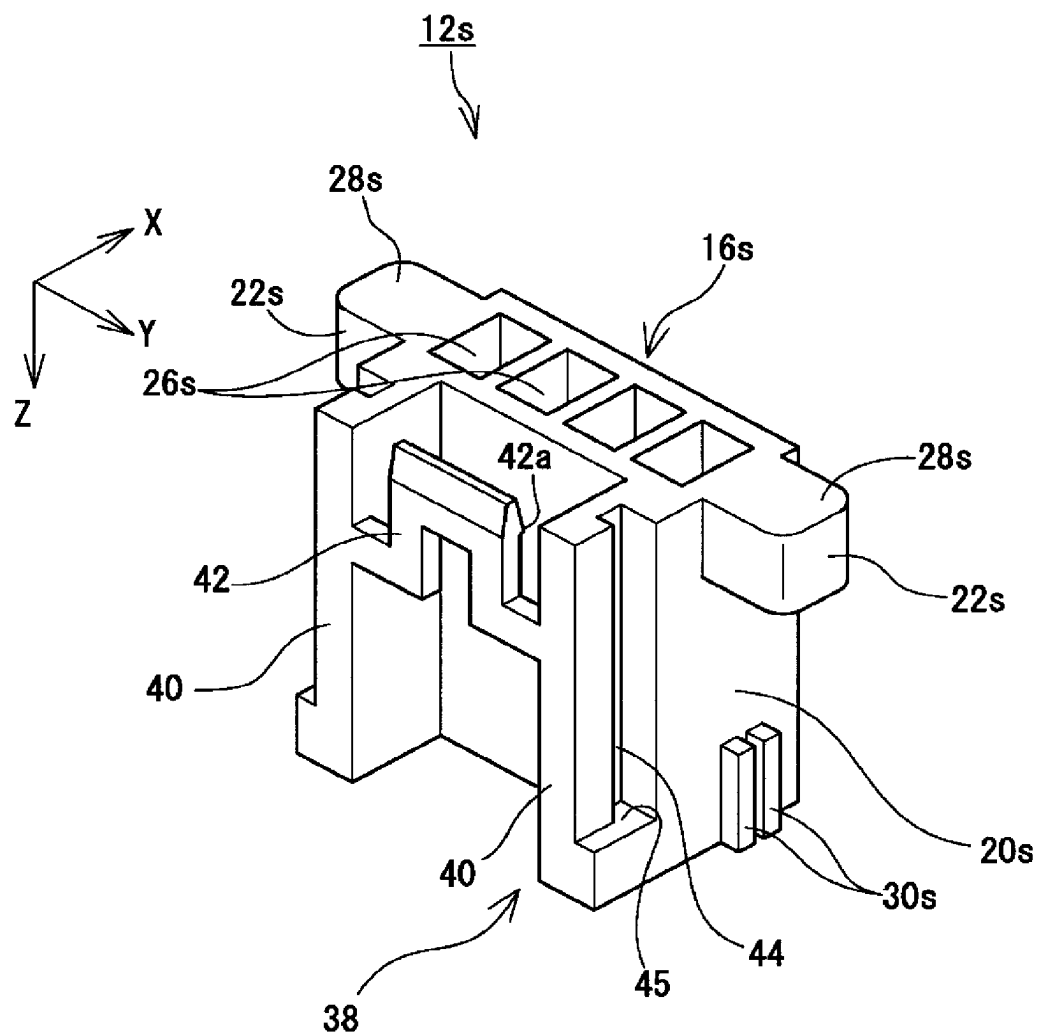


FIG. 4

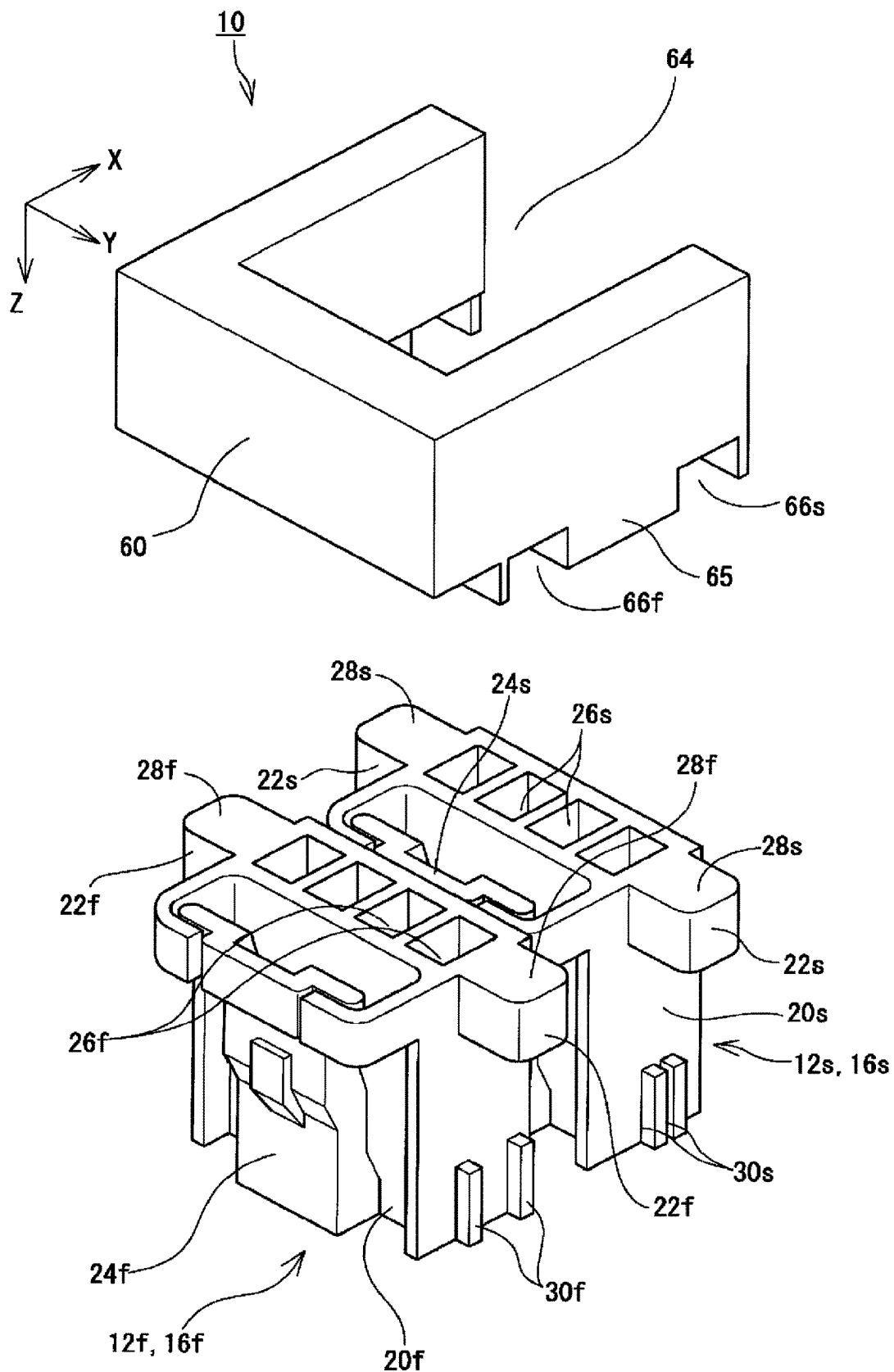


FIG. 5

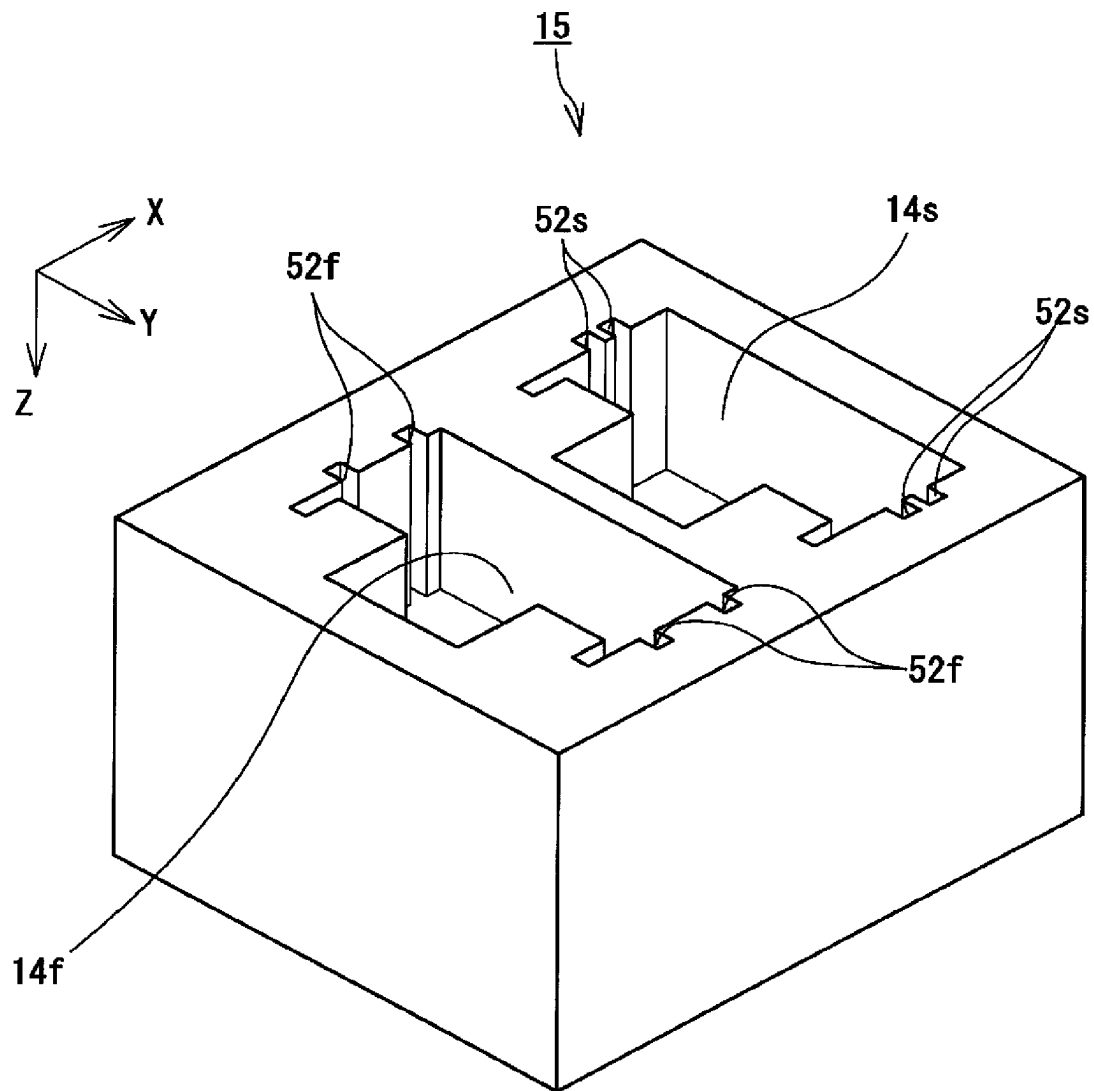


FIG. 6

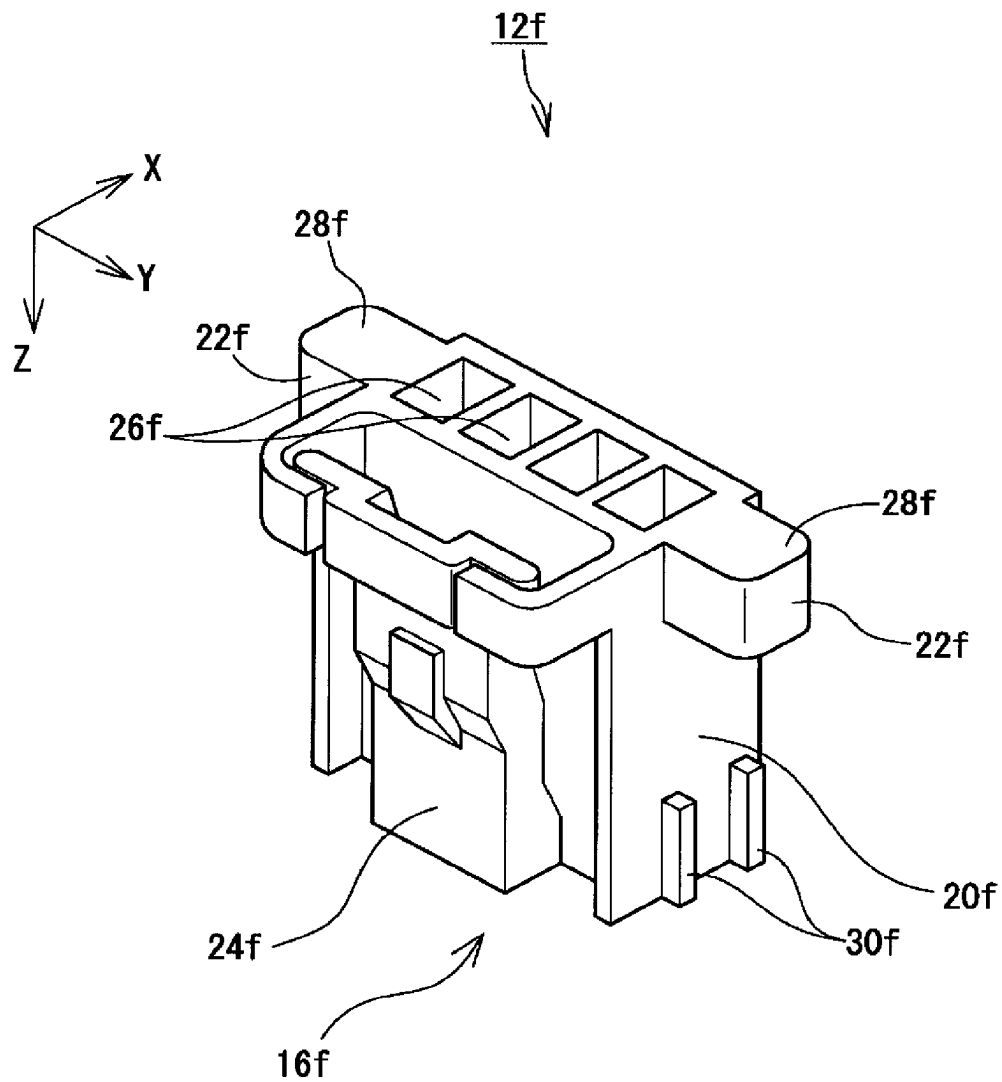


FIG. 7

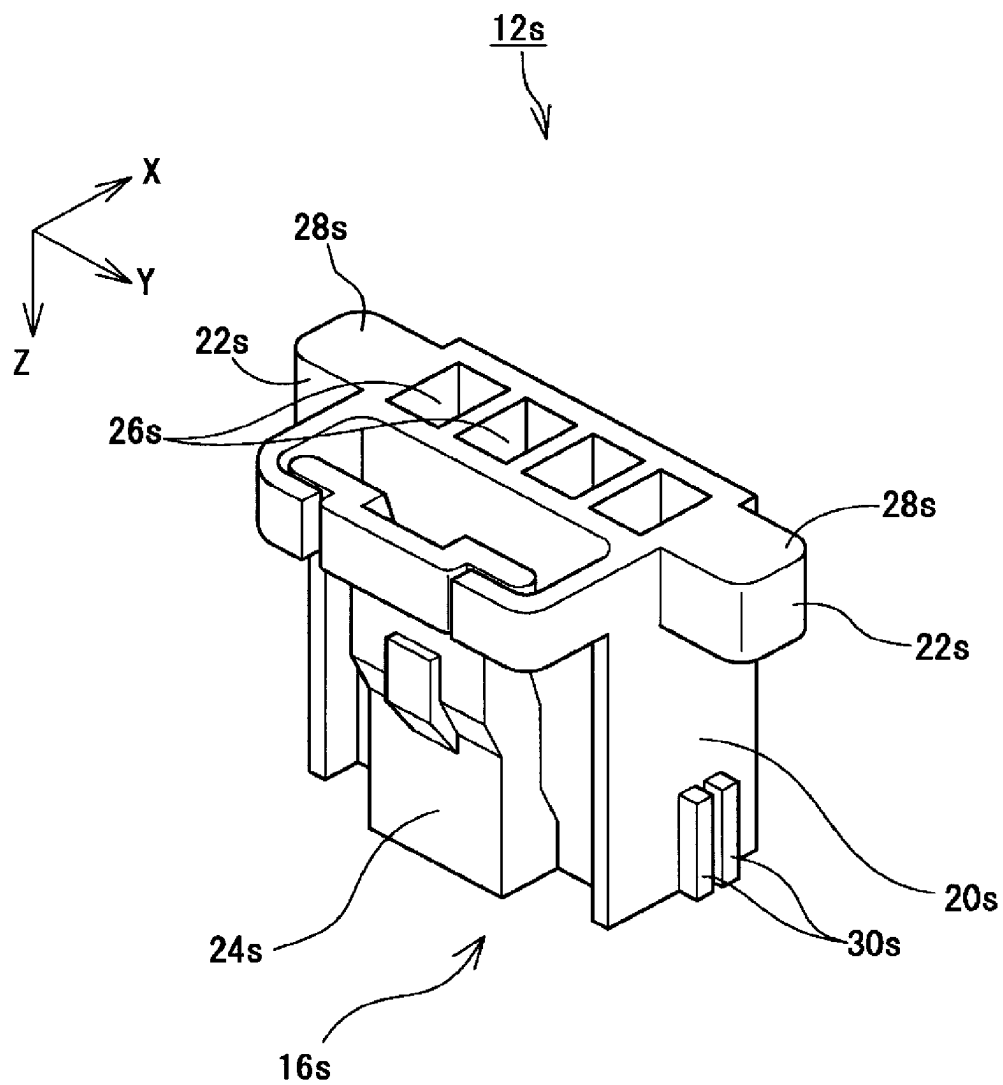


FIG. 8

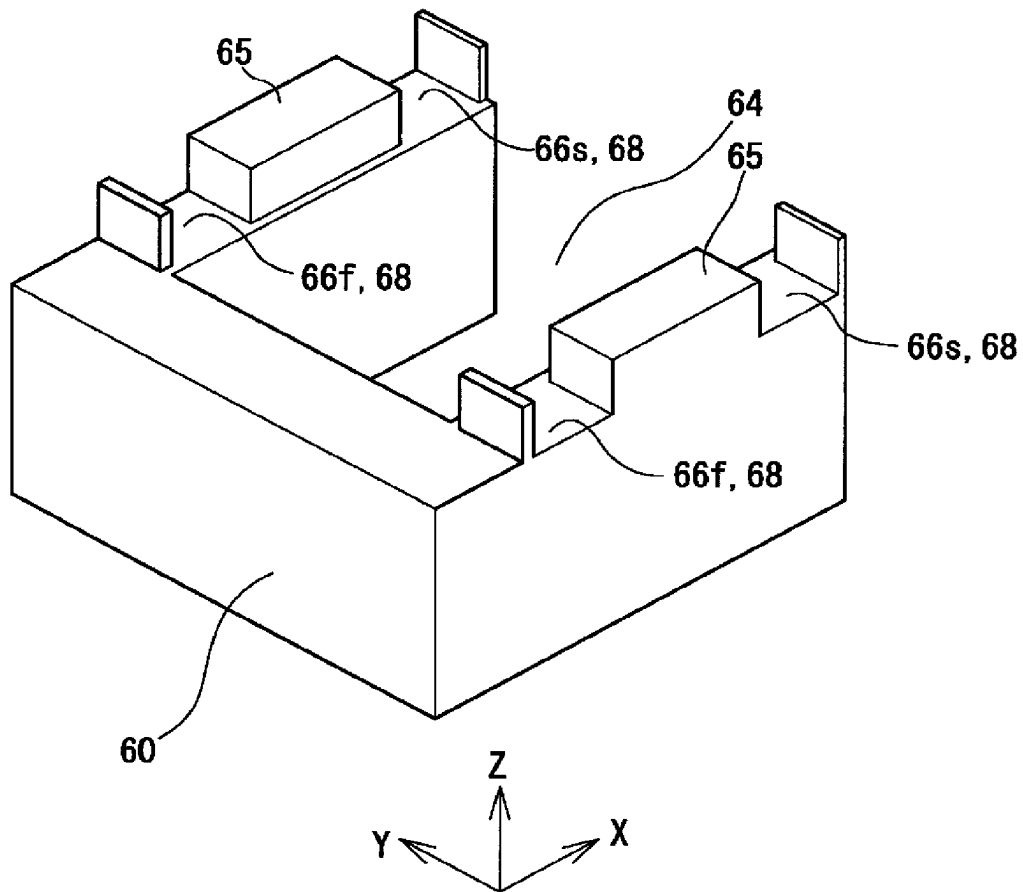


FIG. 9

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CONNECTOR SET MATING METHOD FOR CONNECTING TWO PLUG CONNECTORS TO TWO SOCKET CONNECTORS AND A CONNECTOR SET USING THE SAME

CROSS REFERENCE TO RELATED APPLICATION

This application claims priority to Japanese Patent Application No. 2021-046094 filed on Mar. 19, 2021, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

The present specification discloses a connector mating method for respectively inserting and mating a first plug connector into a first receptacle connector and a second plug connector into a second receptacle connector, and also discloses a connector set comprising the first plug connector and the second plug connector.

BACKGROUND

Typically, connector devices include a receptacle connector having a recess for connection and a plug connector having a protrusion to be inserted into the recess of the receptacle connector. Some plug connectors retain a plurality of electric wires aligned along a predetermined arrangement direction.

For example, JP 2019-87408 A (Patent Document 1) discloses a plug connector (which is referred to as a “first connector” in Patent Document 1) equipped with a housing for retaining a plurality of electric wires which are aligned along a predetermined arrangement direction. It has been desired that the size of such plug connectors be further minimized. In particular, when a plug connector is utilized in a situation where a great number of electronic devices are arranged in a limited space, as in the case of a vehicle, it is desirable to further minimize the plug connector.

CITATION LIST

Patent Literature

Patent Document 1: JP 2019-87408 A

Under the circumstances, attempts to decrease dimensions of plug connectors have been made conventionally. For example, in the plug connector of Patent Document 1, its dimensions along a thickness direction orthogonal to the arrangement direction are reduced as much as possible. In the above-described configuration, however, a region which is pushed by fingers in order to mate the plug connector with the receptacle connector is limited to only two points located on both sides of a plurality of electric wires in the arrangement direction. When only the two points are pushed as described above, the plug connector is easily inclined about a line passing through the two points. As a result, it has been difficult to appropriately insert the plug connector into the receptacle connector.

In other words, minimization of the plug connector conventionally results in poor workability of operation to insert the plug connector into the receptacle connector.

In view of the circumstances, the present specification discloses a connector mating method and a connector set with which workability in operation of inserting a plug

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connector into a receptacle connector can be improved while maintaining the plug connector in a thin shape.

SUMMARY

A connector mating method disclosed in this specification is a method for inserting a first plug connector into a first receptacle connector and a second plug connector into a second receptacle connector to mate the connectors, in which each of the first plug connector and the second plug connector includes a main body configured to retain a plurality of electric wires and a pair of push members protruding from both sides of the main body along a first direction, and in which the first plug connector and the second plug connector are mechanically coupled to each other or bound by a jig to a relative positional relationship therebetween in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction, and are inserted into the first plug connector and the second plug connector while being mechanically coupled to each other or being bound to the relative positional relationship by the jig.

In the above-described configuration, a total of four push members disposed on the first and second plug connectors are pushed while maintaining a positionally restricted relationship therebetween. This makes the first and second plug connectors less prone to inclination. As a result, workability in operation to insert the plug connector into the receptacle connector can be improved while maintaining a small size of each of the plug connectors.

In this case, a plurality of wire connection holes through which the electric wires are inserted may be arranged in a line along the first direction.

When the plurality of wire connection holes are arranged in the line along the same direction as an arrangement direction in which the pair of push members are arranged (i.e., the first direction), the plug connector can be configured to have smaller dimensions in the second direction, and accordingly have a thin outline.

In an aspect of this disclosure, the first plug connector and the second plug connector may be mechanically coupled to each other or bound by the jig to the relative positional relationship in a position where the first plug connector and the second plug connector are adjacent to each other in the second direction.

When configured as described above, the first and second plug connectors mated with the first and second receptacle connectors can be made compact in size.

In an aspect of the present disclosure, each of the first plug connector and the second plug connector may further include a guide rib extending along an insertion direction of the plug connectors, the guide rib being designed to be received in a guide groove defined in an associated one of the receptacle connectors. Further, a relative position between the guide rib and the main body in the first plug connector may be varied from a relative position between the guide rib and the main body in the second plug connector, and when one of the plug connectors tries to enter an unassociated one of the receptacle connectors, the guide rib interferes with a part of the unassociated one of the receptacle connectors to thereby block further entry of the plug connector.

The above-described configuration can reliably prevent an insertion error where the plug connector is inserted into an unassociated one of the receptacle connectors.

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A connector set disclosed herein includes a first plug connector configured to be inserted into a first receptacle connector and a second plug connector configured to be inserted into a second receptacle connector, in which each of the first plug connector and the second plug connector has a main body configured to retain a plurality of electric wires and a pair of push members protruding along a first direction from both sides of the main body. Further, in the connector set, the first plug connector is mechanically connectable to the second plug connector in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction.

When configured as described above, a total of four push members disposed on the first and second plug connectors are pushed while maintaining the positionally restricted relationship therebetween. This makes the first and second plug connectors less prone to inclination about a line. As a result, workability in operation to insert the plug connectors into the receptacle connectors can be improved while maintaining a small size of each of the plug connectors.

Another connector set disclosed herein includes a first plug connector configured to be inserted into a first receptacle connector, a second plug connector configured to be inserted into a second receptacle connector, and a jig, in which each of the first plug connector and the second plug connector includes a main body configured to retain a plurality of electric wires and a pair of push members protruding along a first direction from both sides of the main body. Further, in the other connector set, the jig includes a binding member which is configured to house a part of each of the first and second plug connectors in such a manner that the first and second plug connectors are bound to a relative positional relationship therebetween in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction.

When configured as described above, a total of four push members disposed on the first and second plug connectors are pushed while maintaining the positionally restrained relationship therebetween. This makes the first and second plug connectors less prone to inclination about a line. In this way, workability in operation to insert the plug connectors into the receptacle connectors can be improved while maintaining a small size of each of the plug connectors.

A technique disclosed herein can ensure that the workability in operation to insert the plug connectors into the receptacle connectors is further improved while maintaining the small size of each of the plug connectors.

BRIEF DESCRIPTION OF DRAWINGS

Embodiments of the present disclosure will be described based on the following figures, wherein:

FIG. 1 is a perspective view of a connector set;
FIG. 2 is a perspective view of a terminal base;
FIG. 3 is a perspective view of a first plug connector;
FIG. 4 is a perspective view of a second plug connector;
FIG. 5 is a perspective view of another connector set;
FIG. 6 is a perspective view of another terminal base;
FIG. 7 is a perspective view of a first plug connector in the connector set shown in FIG. 5;

FIG. 8 is a perspective view of a second plug connector in the connector set shown in FIG. 5; and

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FIG. 9 is a perspective view of a jig which is flipped upside down from a state shown in FIG. 5.

DESCRIPTION OF EMBODIMENTS

Hereinafter, a connector set 10 incorporating two plug connectors composed of a first plug connector 12_f and a second plug connector 12_s, and a connector mating method using the connector set 10 will be described with reference to the drawings. FIG. 1 shows a perspective view of the connector set 10, and FIG. 2 shows a perspective view of a terminal base 15 which has two receptacle connectors 14_f and 14_s to receive the connector set 10. FIGS. 3 and 4 respectively show perspective views of the first plug connector 12_f and the second plug connector 12_s. In the following description, a direction in which the plug connectors 12_f and 12_s are inserted into the receptacle connectors 14_f and 14_s is referred to as an "insertion direction", a direction in which a pair of push members 22, which will be described below, are arranged is referred to as a "first direction", and a direction orthogonal to both the insertion direction and the first direction is referred to as a "second direction". Further, in the following description, a posterior part of the insertion direction is considered a "downside", and an anterior part of the insertion direction is considered an "upside". However, the terms "downside" and "upside" are irrelevant to gravitational downside and upside. In the drawings, the "first direction", the "second direction", and the "insertion direction" are described as "Y", "X", and "Z", respectively.

The connector set 10 includes two plug connectors consisting of the first plug connector 12_f and the second plug connector 12_s. The terminal base 15 has a first receptacle connector 14_f in which the first plug connector 12 is inserted and a second receptacle connector 14_s in which the second plug connector 12_s is inserted. In this example, the two receptacle connectors 14_f and 14_s are merged with each other to form one large recess. Each of the plug connectors 12_f and 12_s is inserted into an associated one of the receptacle connectors 14_f and 14_s and fitted therein to electrically connect on-board electronic devices installed in a vehicle to other on-board electronic devices.

More specifically, the first plug connector 12_f is connected to a first temperature sensor (not illustrated) via four electric wires 18_f (in FIG. 1, only one of the electric wires 18_f is illustrated, and is omitted in FIG. 3). The second plug connector 12_s is connected to a second temperature sensor (not illustrated) via four electric wires 18_s (in FIG. 1, only one of the electric wires 18_s is illustrated, and is omitted in FIG. 3). The first temperature sensor is installed in a not-illustrated first rotating electric machine which is mounted on the vehicle to detect the temperature of the first rotating electric machine. Similarly, the second temperature sensor is incorporated in a not-illustrated second rotating electric machine which is mounted on the vehicle to detect the temperature of the second rotating electric machine. When the plug connectors 12_f and 12_s are inserted into the receptacle connectors 14_f and 14_s, the first and second temperature sensors are electrically connected to other electronic devices and become able to transmit and receive electric power and electric signals.

The first plug connector 12_f and the second plug connector 12_s, which are completely separated independent parts, are mechanically coupled and merged into one unit as shown in FIG. 1 when the plug connectors 12_f and 12_s are inserted into the corresponding receptacle connectors 14, respec-

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tively. Configurations of the first plug connector 12f and the second plug connector 12s are explained in detail below.

The first plug connector 12f has a resin housing 16f and an electric mechanism (not illustrated) arranged within the resin housing 16f. The electric mechanism includes components, such as an input-side contact to be brought into contact with the electric wire 18f, an output-side contact contactable with a contact of the first receptacle connector 14f, and a connection line for electrically connecting the input-side contact and the output-side contact. As the components can be implemented by conventionally known techniques, details of the components are not explained herein.

The housing 16f is roughly divided into five portions consisting of a main body 20f, a push member 22f, a retainer lug 24, a guide rib 30, and a coupler member 32. The main body 20f is an approximately rectangular member configured to retain the electric mechanism and a plurality of electric wires 18f. An upper end surface of the main body 20f located on an upper side in the insertion direction (Z direction) is provided with four wire connection holes 26f which are arranged in a row along the first direction (Y direction). The electric wires 18f are respectively inserted into the wire connection hole 26f. The electric wires 18f inserted into the wire connection holes 26f are secured in a state of being in contact with the input-side contact. Dimensions of the main body 20f in the second direction (X direction) and the first direction are defined as small as possible within a range capable of maintaining appropriate strength. As a result, the upper end surface of the main body 20f is in most areas occupied by the wire connection holes 26f, leaving almost no area for fingers to push the main body 20f on the upper end surface.

The retainer lug 24 is fixed to an end surface of the main body 20 that is located at one of second direction ends of the main body 20. The retainer lug 24 is a cantilever arm type member which is fixed only at its lower portion to the main body 20f, and is configured to be swingable through flexure in the second direction. When the first plug connector 12f is inserted into the first receptacle connector 14f, a part of the retainer lug 24 is engaged with a part of the first receptacle connector 14f. This can prevent unintended removal of the first plug connector 12f from the first receptacle connector 14f. The first plug connector 12f can be withdrawn and removed from the first receptacle connector 14f by pulling the first plug connector 12f upward in a state where the retainer lug 24 is swung toward the main body 20f.

A pair of push members 22f are protruded outward along the first direction from side surfaces of the main body 20f located at both first direction ends thereof. Upper end surfaces of the push members 22f function as pushing surfaces 28f which are pushed by fingers in order to insert the first plug connector 12f into the first receptacle connector 14f. The pushing surfaces 28f are flush with the upper end surface of the main body 20f, i.e. the surface in which the wire connection holes 26f are defined.

The coupler member 32 is designed to be engaged with a part of the second plug connector 12s for coupling the first plug connector 12f to the second plug connector 12s. The coupler member 32 includes a coupling loop 34 (see FIG. 3) and a plurality of slide claws 36. It should be noted that, in FIG. 3, the coupling loop 34 is located at an invisible position, and is therefore indicated by a broken line in the drawing.

The coupling loop 34 is outwardly protruded along the second direction from an end surface of the main body 20f that is located at the other of the second direction ends of the main body 20f (i.e., the end surface opposite to the retainer

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lug 24), and is configured to define a vertical through hole between the end surface of the main body 20f and the coupling loop 34 itself. A coupling lug 42 in the second plug connector 12s, which will be described below, enters the hole to engage with the coupling loop 34.

The slide claws 36 are outwardly protruded along the second direction, at both first direction ends, from the end surface of the main body 20f that is located at the other of the second direction ends of the main body 20f. A total of four slide claws 36 are arranged in a two-by-two matrix in which the slide claws 36 are spaced from each other in both the first direction and the vertical direction. Each of the slide claws 36 is formed in an approximately L-letter shape with its tip end being bent inward in the first direction. When such tip ends of the slide claws 36 are fitted in below-described slide rails 44 of the second plug connector 12s, the first plug connector 12f is allowed to move relative to the second plug connector 12s along the insertion direction, while being restricted so as not to move along the first direction or the second direction.

Guide ribs 30f extending along the insertion direction are formed on both end surfaces of the main body 20f that are located at first direction ends of the main body 20f. In this example, two guide ribs 30f are disposed for each of the end surfaces, and a total of four guide ribs 30 are provided to the entire first plug connector 12f. The first receptacle connector 14f has guide grooves 52f which are configured to receive the guide ribs 30f. When the guide ribs 30f are inserted into the guide grooves 52f, movement of the first plug connector 12f is guided in the vertical direction (i.e., the insertion direction).

Next a configuration of the second plug connector 12s will be described. The second plug connector 12s includes, as in the case of the first plug connector 12f, a resin housing 16s and a not-illustrated electric mechanism installed within the housing 16s. The housing 16f includes a main body 20s, push members 22s, a coupling member 38, and guide ribs 30s. The main body 20s and the push members 22s are almost identical in structure to the main body 20f and the push members 22f of the first plug connector 12f, and explanation related to these components is not repeated.

The coupling member 38 includes a pair of rail members 40 and the coupling lug 42. The rail members 40 are formed in a flat plate shape extending outward in the second direction from an end surface of the main body 20s that is located on one of second direction ends of the main body 20s. The pair of rail members 40 are spaced from each other in the first direction. A slide rail 44 extending along the vertical direction is formed on an outer end surface facing outward in the first direction for each of the rail members 40. The slide rail 44 extends, as shown in FIG. 4, penetrating through an upper end surface of the rail member 40. Therefore, the tip end of the slide claw 36 in the first plug connector 12f can be inserted from above the second plug connector 12s into the slide rail 44. On the other hand, the slide rail 44 does not reach a lower end of the rail member 40, and a stopper surface 45, which is a flat plane facing upward, is defined in the rail member 40 at a height that matches a lower end of the slide rail 44. The stopper surface 45 blocks the slide claw 36 to prevent the first plug connector 12f from sliding further below the stopper surface 45.

The coupling lug 42 is disposed between the pair of rail members 40. The coupling lug 42 is a lug member protruding upward. The coupling lug 42 has a hitching portion 42a protruding along the second direction in the vicinity of a tip end of the coupling lug 42, and an upper area of the coupling lug 42 located above the hitching portion 42a is formed in

a tapered shape whose dimension in the second direction is gradually decreased toward the top of the coupling lug 42.

Similarly to the guide ribs 30f, the guide ribs 30s extending along the insertion direction are formed on both end surfaces of the main body 20s that are located on first direction ends of the main body 20s. Guide grooves 52s for receiving the guide ribs 30s are defined in the second receptacle connector 14s.

When the first plug connector 12f and the second plug connector 12s are inserted into the first receptacle connector 14f and the second receptacle connector 14s, the first plug connector 12f and the second plug connector 12s are coupled and merged into one unit. Specifically, in operation to couple the plug connectors 12f and 12s, the slide claws 35 are fitted into the slide rails 44 from above the side rails 44, and the first plug connector 12f is slid downward on the second plug connector 12s. The downward sliding of the first plug connector 12f is guided by the slide rails 44. The coupling lug 42 enters the hole in the coupling loop 34 at some point during the sliding of the first plug connector 12f.

Then, the first plug connector 12f is slid further downward until the slide claws 36 located on a lower part make contact with the stopper surface 45. At this time, the hitching portion 42a passes through the coupling loop 34 to surmount the coupling loop 34. The contact of the lower slide claws 36 with the stopper surface 45 hampers further downward movement of the first plug connector 12f. Meanwhile, when the hitching portion 42a has surmounted the coupling loop 34, the hitching portion 42a and the coupling loop 34 are engaged with each other, to thereby hamper upward movement of the first plug connector 12f. In other words, when the lower slide claws 34 are brought into contact with the stopper surface 45, the first plug connector 12f becomes unable to move relative to the second plug connector 12s, and the first and second plug connectors 12f and 12s are mechanically coupled to each other accordingly.

Dimensions of each section in the coupler member 32 and the coupling member 34 are defined in such a manner that the two plug connectors 12f and 12s are aligned at the same height when coupling between the plug connectors 12f and 12s is complete. Therefore, in this example, a total of four pushing surfaces 28f, 28s are located on the same plane when the two plug connectors 12f and 12s are completely coupled to each other. In addition, when the coupling is complete, the pair of pushing surfaces 28f in the first plug connector 12f are displaced in the second direction from the pair of pushing surfaces 28s in the second plug connector 12s.

After coupling the two plug connectors 12f and 12s, an operator inserts the two coupled plug connectors 12f and 12s into the hole in the receptacle connectors 14f and 14s. In this inserting operation, the operator pushes the four pushing surfaces 28f and 28s from thereabove with their hands. Then, mating of the connectors is finally complete when the plug connectors 12f and 12s are inserted into the receptacle connectors 14f and 14s to a position where the retainer lug 24 is engaged with a part of the first receptacle connector 14f.

As can be clearly understood from the above description, the two plug connectors 12f and 12s are mechanically coupled to each other in order to insert the two plug connectors 12f and 12s into the receptacle connectors 14f and 14s in this example. The reason for mechanically coupling the two plug connectors 12f and 12s is described below.

As described above, the first plug connector 12f in this example is connected to the first temperature sensor, and the

second plug connector 12s is connected to the second temperature sensor. Further, the first temperature sensor is incorporated in the first rotating electric machine, while the second temperature sensor is incorporated in the second rotating electric machine. Both the first rotating electric machine and the second rotating electric machine are mounted on the vehicle. Here, in addition to the rotating electric machines, a variety of electronic devices are mounted on the vehicle, while a space available for mounting such components within the vehicle is limited. For this reason, it is desired that on-board components including the plug connectors 12f and 12s be minimized to the greatest possible extent.

To satisfy such a desire, the main bodies 20f and 20s of the plug connectors 12f and 12s are reduced in size to a maximum possible extent at which the strength of the main bodies 20f and 20s can be properly maintained. As a result, the upper end surfaces of the main bodies 20f and 20s are mostly occupied by the wire connection holes 26, leaving almost no areas available for pushing by fingers. Therefore, in this example, the push members 22 are provided at both first direction ends of the main bodies 20f and 20s to secure the pushing surfaces 28f and 28s which can be pushed by the fingers.

Here, it is assumed that the first plug connector 12f is solely inserted into the first receptacle connector 14f without being mechanically coupled to the first plug connector 12s. In this case, the operator pushes the two pushing surfaces 28f located on both sides of the main body 20f by hand. In other words, only two points on a line along the first direction are pushed by the operator. When pushed in this way, the first plug connector 12f is apt to sway about the line passing through the two points. As a result, there is a risk that the first plug connector 12f may be inserted in a state of being inclined relative to the first receptacle connector 14f. When the first plug connector 12f is inclined, the guide ribs 30f butt against the guide grooves 52f, which hinders further insertion of the first plug connector 12f. Therefore, it has been necessary for the operator to slightly adjust the first plug connector 12f to correct for the position every time the first plug connector 12f is inclined. That is, in a case where the first plug connector 12f is solely inserted into the first receptacle connector 14f, because the operator is repeatedly required to adjust the position of the first plug connector 12f which is apt to be inclined, connector mating work can often be time consuming.

With this in mind, in connector mating operation according to this example, the first plug connector 12f and the second plug connector 12s are mechanically coupled to be merged into one unit. As a result, the merged plug connectors 12f and 12s are equipped with the four pushing surfaces 28f and 28s as a whole. The four pushing surfaces 28f and 28s are spaced from each other in both the first direction and the second direction, and are not aligned in a row. In an action of pushing the four pushing points, inclination about a line that passes through two pushing points among the four pushing points can be prevented by a force applied to the other two pushing points. For example, in FIG. 1, if the first plug connector 12f sways around a line L1 passing through the two pushing surfaces 28f, the pushing surfaces 28s attempt to move upward. However, when the pushing surfaces 28s are pushed simultaneously with the pushing surfaces 28f, the first plug connector 12f is not caused to sway. Similarly, in a case where the first plug connector 12f attempts to sway around a line L2 passing through the pushing surface 28f and the pushing surface 28s which are located on the right side in FIG. 1, the attempt to sway can

be prevented by a downward force applied to the pushing surface **28f** and the pushing surface **28s** which are located on the left side in FIG. 1. This can eliminate the necessity for the operator to repeatedly adjust the position of the plug connectors **12f** and **12s** during operation to insert the plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s**, and can thus improve workability of mating the connectors.

Meanwhile, for securing four push points, it is also conceivable that the first plug connector **12f** and the second plug connector **12s** are initially formed as one integral connector rather than being formed as mechanically couplable connectors. That is, it may be considered to prepare a single plug connector **12** including a single main body **20** in which eight wire connection holes **26** are defined, four push members **22** protruded from the single main body **20**, and a single retainer lug **24** fixed to the single main body **20**. Even with this configuration of the plug connector **12**, the four pushing surfaces **28** can be secured for pushing the plug connector **12**, to thereby prevent inclination thereof.

However, in this case, workability in assembling the temperature sensor or the rotating electric machine will be deteriorated. Specifically, the first and second temperature sensors are respectively connected to the first and second plug connectors **12f** and **12s**, and respectively attached, in the connected state, to the first and second rotating electric machines. Further, the first and second rotating electric machines incorporating the first and second temperature sensors are installed in the vehicle. In a case where the first plug connector **12f** and the second plug connector **12s** are integrated into one plug connector **12**, the first temperature sensor has been mechanically coupled to the second temperature sensor via the plug connector **12** and the electric wires **18**. In this case, an action of moving one of the first and second temperature sensors is transferred to the other of the first and second temperature sensors, which often causes unintentional behavior of the temperature sensors. As a result, the operator is not able to move the temperature sensors and the rotating electric machines as desired, resulting in deteriorated workability in installation.

With this in mind, in this example, the first plug connector **12f** and the second plug connector **12s** are formed as completely separated independent components, and mechanically coupled to each other only when the connectors are mated. In this way, it becomes possible to improve workability in mating the connectors while improving workability in installing the temperature sensors and the rotating electric machines.

Next, another connector set **10** and another connector mating method are explained with reference to FIGS. **5** to **9**. FIG. **5** shows a perspective view of the connector set **10**, and FIG. **6** shows a perspective view of a terminal base **15**. FIG. **7** shows a perspective view of a first plug connector **12f**, and FIG. **8** shows a perspective view of a second plug connector **12s**. In addition, FIG. **9** shows a perspective view of a jig **60** which is flipped upside down from a state shown in FIG. **5**.

The connector set **10** shown in FIG. **5** includes the jig **60** in addition to the first plug connector **12f** and the second plug connector **12s**. The jig **60** is designed to bind a relative positional relationship between the first plug connector **12f** and the second plug connector **12**, which will be described further below.

The first plug connector **12f** includes the main body **20f**, the push member **22f**, the retainer lug **24f**, and the guide ribs **30f**. Configurations of the main body **20f**, the push member **22f**, the retainer lug **24**, and the guide ribs **30f** are almost identical to those of the first plug connector **12f** shown in

FIGS. **1** and **3**, and the descriptions related to the configurations are not repeated. It should be noted that the first plug connector **12f** in this example does not include the coupler member **32**, in contrast to the first plug connector **12f** shown in FIGS. **1** and **3**.

The second plug connector **12s** has a configuration almost identical to the first plug connector **12f**, and includes the main body **20s**, the push member **22s**, the retainer lug **24s**, and the guide ribs **30s**. However, the guide ribs **30s** are arranged at positions shifted toward the center in the second direction relative to the guide ribs **30f**. That is, a positional relationship between the guide ribs **30s** and the main body **20s** is different from a positional relationship between the guide ribs **30f** and the main body **20f**.

The terminal base **15** shown in FIG. **6** has a first receptacle connector **14f** configured to receive the first plug connector **12f** and a second receptacle connector **14s** configured to receive the second plug connector **12s**. The first receptacle connector **14f** and the second receptacle connector **14s** have a very similar formation, yet have a difference in position between guide grooves **52f** and guide grooves **52s** that are configured to receive the guide ribs **30f** and **30s**, respectively.

For this reason, when it is attempted to insert the first plug connector **12f** into the second receptacle connector **14s**, the guide ribs **30f** interfere with a part of the second receptacle connector **14s**. Similarly, when it is attempted to insert the second plug connector **12s** into the first receptacle connector **14f**, the guide ribs **30s** interfere with a part of the first receptacle connector **14f**. In other words, an insertion error where the plug connector **12f** or **12s** is inserted into an unassociated one of the receptacle connectors **14s** and **14f** can be reliably prevented by varying the positional relationship between the guide ribs **30f** and the main body **20f** from the positional relationship between the guide ribs **30s** and the main body **20s**.

When inserting the plug connectors **12f** and **12s** into the receptacle connectors **14f** and **14s**, the operator binds a relative positional relationship between the plug connectors **12f** and **12s** by the jig **60**. The jig **60** is a substantially U-letter shaped member which is open at one of its second direction ends when viewed from above. In other words, the jig **60** is open to the outside at the one of the second direction ends and has a cutout space **64** penetrating through the jig **60** along the vertical direction. The cutout space **64** is defined to allow the electric wires **18f** and **18s** to pass therethrough.

In addition, a pair of binding walls **65** are downwardly protruded from a lower end surface of the jig **60**. The pair of binding walls **65** are spaced from each other along the first direction, and a spacing distance in the first direction between the two binding walls **65** is substantially equal to or slightly greater than a dimension, in the first direction, of the main body **20**. In addition, each of the binding walls **65** has two binding grooves **66f** and **66s** which are spaced from each other in the second direction and configured to receive the push members **22f** and **22s**, respectively. Accordingly, in the entire jig **60**, a total of four binding grooves **66f** and **66s** are formed.

The binding grooves **66f** and **66s** respectively receive partial regions (i.e., the push members **22f** and **22s**) in the two plug connectors **12f** and **12s** being adjacently aligned along the second direction, to thereby function as a binding member for binding the relative positional relationship between the two plug connectors **12f** and **12s**. Dimensions, in the second direction, of the binding grooves **66f** and **66s** are defined to be approximately equal to or slightly smaller than dimensions in the second direction of the push mem-

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bers 22f and 22s. In this connection, as shown in FIGS. 5 and 9, walls of the binding grooves 66f and 66s located at second direction ends of each of the binding walls 65 are relatively thin and bendable like a leaf spring. When the push members 22f and 22s are respectively press fitted into the binding grooves 66f and 66s, the push members 22f and 22s, and thus the plug connectors 12f and 12s, are retained due to a frictional force or an elastic restoring force of the leaf spring by the jig 60. Further, bottom surfaces of the binding grooves 66 function as contact surfaces 68 for pushing the pushing surfaces 28.

In operation to mate the connectors, the operator positions the main bodies 20f and 20s between the pair of the binding walls 65 and then forces the push members 22f and 22s to enter the binding grooves 66f and 66s. As a result, the pair of binding walls 65 inhibit movement of the plug connectors 12f and 12s in the first direction while the binding grooves 66f and 66s inhibit movement of the plug connectors 12f and 12s in the second direction. In addition, the plug connectors 12f and 12s are retained within the jig 60 by the frictional force occurring between the binding groove 66f, 66s and the push member 22f, 22s or the elastic restoring force of the leaf spring, without escaping downward from the jig 60. As a result, the two plug connectors 12f and 12s are bound to a positional relationship in which the plug connectors 12f and 12s are adjacent to each other in the second direction.

In this state, the operator places the plug connectors 12f and 12s in position relative to the associated receptacle connectors 14f and 14s and pushes the jig 60 from above. This causes the plug connectors 12f and 12s to respectively enter the associated receptacle connectors 14f and 14s. In this operation, the four pushing surfaces 28 are simultaneously pushed by the contact surfaces 68 in the jig 60. Here, a top surface of the jig 60 has a sufficient space to be held by a hand or fingers, which allows the operator to planarly push and press the jig 60. It is therefore possible to downwardly push the plug connectors 12f and 12s while preventing the jig 60, and thus the plug connectors 12f and 12s, from getting inclined. As a result, there is no necessity to constantly adjust the position of the plug connectors 12f and 12s during operation to insert the plug connectors 12f and 12s into the receptacle connectors 14f and 14s, and workability in operation to mate the connectors can be accordingly improved.

It should be noted that the four binding grooves 66f and 66s, all of which have the same shape, might invite a fitting error which is caused by incorrectly fitting the push member 22s into the binding groove 66f/being essentially intended to receive the push member 22f. However, if an attempt to insert the plug connectors 12f and 12s into the receptacle connectors 14f and 14s is made in such an erroneously fitted state, the guide ribs 30f and 30s will interfere with a part of the receptacle connectors 14f and 14s, and accordingly prevent the attempt to insert the plug connectors 12f and 12s from being successfully accomplished. In this way, an insertion error of the plug connectors 12f and 12s can be reliably prevented according to this example.

When the two plug connectors 12f and 12s enter the receptacle connectors 14f and 14s to the deepest position, the retainer lug 24 is locally engaged with the receptacle connectors 14f and 14s. After the engagement is established, the operator pulls up the jig 60 to remove the jig 60 from the plug connectors 12f and 12s. Here, because the jig 60 has a holding force which is sufficiently smaller than an engagement force exerted by the retainer lug 24, it is possible to remove the jig 60 from the plug connectors 12f and 14s only

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by lifting the jig 60. When the jig 60 is removed in this way, the connector mating work is complete.

As can be clearly understood from the above explanation, the relative positional relationship between the two plug connectors 12f and 12s is bound by the jig 60, and the thus-bound plug connectors 12f and 12s can be planarly pushed and pressed from the top of the jig 60. As a result, the plug connectors 12f and 12s can be effectively prevented from getting inclined, which makes the work to insert the plug connectors 12f and 12s into the receptacle connectors 14f and 14s quick and easy.

The above-described configuration is presented by way of illustration, and may be modified in various ways other than features that each of the plug connectors 12f and 12s has the pair of push members 22f, 22s protruding along the first direction from both sides of the main body 20f, 20s, and that the two plug connectors 12f and 12s are mechanically coupled to each other or bound by the jig 60 in a position where the push members 22f are displaced in the second direction from the push members 22s, and are inserted into the associated receptacle connectors 14f and 14s while being mechanically coupled to each other or bound by the jig 60.

For example, the shapes and other features of the plug connectors 12f and 12s may be changed as appropriate. For example, the second plug connector 12s, which does not include the retainer lug 24 in the example shown in FIG. 1, may have a retainer lug 24 on a surface located opposite the coupling member 38. Further, in the example of FIG. 1, the way of coupling the two plug connectors 12f and 12s may be changed as appropriate. For example, the two plug connectors 12f and 12s may be coupled to each other by a magnetic force or a hook-and-loop fastener, by press fitting, or by other component. Also in the example shown in FIG. 5, the shapes of the plug connectors 12f and 12s and the jig 60 may be changed as appropriate.

In the above description, the connector set 10 includes the two plug connectors 12f and 12s, but may be configured to include three or more plug connectors 12. For example, the connector set 10 may include three plug connectors which can be mechanically coupled to each other. In addition, the number of electric wires 18f or 18s to be connected to each of the plug connectors 12f or 12s, and arrangement, types, and other features of the electric wires 18f and 18s, may be changed as appropriate. Therefore, the electric wires 18f, 18s may be an electric wire connected to any electronic device other than the temperature sensor.

REFERENCE SIGNS LIST

10 connector set; 12f first plug connector; 12s second plug connector; 14f first receptacle connector; 14s second receptacle connector; 15 terminal base; 16f, 16s housing; 18f, 18s electric wire; 20f, 20s main body; 22f, 22s push member; 24f, 24s retainer lug; 26f, 26s wire connection hole; 28f, 28s pushing surface; 30f, 30s guide rib; 32 coupler member; 34 coupling loop; 36 slide claw; 38 coupling member; 40 rail member; 42 coupling lug; 42a hitching portion; 44 slide rail; 45 stopper surface; 52f, 52s guide groove; 60 jig; 64 cutout space; 65 binding wall; 66f, 66s binding groove; 68 contact surface.

The invention claimed is:

1. A connector mating method for inserting a first plug connector into a first receptacle connector and a second plug connector into a second receptacle connector, to mate the first and second plug connectors to the first and second receptacle connectors,

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wherein each of the first plug connector and the second plug connector comprises
 a main body configured to retain a plurality of electric wires, and
 a pair of push members protruding from both sides of the main body along a first direction, 5
 wherein the first plug connector and the second plug connector are mechanically coupled to each other or bound by a jig to a relative positional relationship therebetween in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction, and inserted into the first receptacle connector and the second receptacle connector while being mechanically coupled to each other or being bound to the relative positional relationship by the jig, 10
 wherein the main body comprises a plurality of wire connection holes in which the plurality of electric wires are respectively inserted, the plurality of wire connection holes and the pair of push members being aligned with each other along the first direction, 15
 wherein the first receptacle connector and the second receptacle connector are recesses that are completely separated from each other, with a partition wall interposed between the first receptacle connector and the second receptacle connector, and 20
 wherein, upon the first plug connector and the second plug connector being mechanically coupled to each other or bound by the jig, a recess for receiving the partition wall is formed between the first plug connector and the second plug connector. 25

2. The connector mating method according to claim 1, wherein the first plug connector and the second plug connector are mechanically coupled to each other or bound to the relative positional relationship by the jig in a position where the first and second plug connectors are adjacent in the second direction. 30

3. The connector mating method according to claim 1, wherein each of the first and second plug connectors further comprises a guide rib extending along an insertion direction, the guide rib being configured to be received in a guide groove defined in an associated one of the first and second receptacle connectors, 35
 wherein a relative position between the guide rib in the first plug connector and the main body is varied from a relative position between the guide rib in the second plug connector and the main body, and 40
 wherein when one of the plug connectors tries to enter an unassociated one of the first and second receptacle connectors, the guide rib interferes with a part of the unassociated one of the first and second receptacle connectors to thereby block further entry of the one of the plug connectors. 45

4. A connector set, comprising: 50
 a first plug connector configured to be inserted into a first receptacle connector; and
 a second plug connector configured to be inserted into a second receptacle connector, 55
 wherein each of the first plug connector and the second plug connector comprises 60

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a main body configured to retain a plurality of electric wires, and
 a pair of push members protruding from both sides of the main body along a first direction,
 wherein the first plug connector and the second plug connector are mechanically couplable to each other in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction,
 wherein the main body comprises a plurality of wire connection holes in which the plurality of electric wires are respectively inserted, the plurality of wire connection holes and the pair of push members being aligned with each other along the first direction,
 wherein the first receptacle connector and the second receptacle connector are recesses that are completely separated from each other, with a partition wall interposed between the first receptacle connector and the second receptacle connector, and
 wherein, upon the first plug connector and the second plug connector being mechanically coupled to each other, a recess for receiving the partition wall is formed between the first plug connector and the second plug connector.

5. A connector set, comprising:
 a first plug connector configured to be inserted into a first receptacle connector;
 a second plug connector configured to be inserted into a second receptacle connector; and
 a jig,
 wherein each of the first plug connector and the second plug connector comprises
 a main body configured to retain a plurality of electric wires, and
 a pair of push members protruding from both sides of the main body along a first direction,
 wherein the jig comprises a binding member configured to partially receive both the first plug connector and the second plug connector for binding a relative positional relationship between the first plug connector and the second plug connector in a position where the pair of push members in the first plug connector are displaced from the pair of push members in the second plug connector along a second direction orthogonal to the first direction,
 wherein the binding member includes a pair of binding grooves that receive two of the push members and a single binding wall defining a respective edge of the pair of binding grooves in the second direction,
 wherein the first receptacle connector and the second receptacle connector are recesses that are completely separated from each other, with a partition wall interposed between the first receptacle connector and the second receptacle connector, and
 wherein, upon the first plug connector and the second plug connector being bound by the jig, a recess for receiving the partition wall is formed between the first plug connector and the second plug connector.

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